

Multi-modal fall detection using improved YOLOv5 and mediapipe

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Abstract

Automated fall detection is critical for elderly care and health monitoring, yet detecting falls accurately remains challenging due to the variability in body orientations, occlusions, and lighting conditions. To address these challenges, this paper presents an enhanced fall detection system based on YOLOv5s and MediaPipe, tailored for improved feature extraction and spatial refinement. Key contributions of our approach include the integration of the Dynamic Atrous Convolution (DAC) and Self-Calibrated Convolution (SCConv) modules into YOLOv5's backbone and head, enhancing multi-scale feature extraction and spatial awareness. The system leverages both MediaPipe's precise keypoint detection and YOLOv5's robust object localization to calculate metrics such as descent velocity and bounding box aspect ratio for reliable fall detection. We evaluate our model using public datasets, demonstrating its robustness across varied scenarios. Experimental results show that our enhanced YOLOv5 model achieves a significant improvement in mean Average Precision (mAP), reaching up to 98.6%, a 4.8% increase from original yolov5s and outperforming conventional methods. The proposed approach offers a promising solution for enhancing the accuracy and reliability of automated fall detection systems, which is crucial for improving elderly healthcare and monitoring.

CCS Concepts

• Computing methodologies; • Artificial intelligence; • Computer vision; • Computer vision problems; • Object detection; • Image and video acquisition; • Motion capture; • Active vision; • Computer vision tasks; • Activity recognition and understanding;

Keywords

YOLOv5s, Dynamic Atrous Convolution (DAC), Self-Calibrated Convolution (SCConv), MediaPipe pose estimation, fall detection

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1 INTRODUCTION

Falls among the elderly pose a significant health risk, often resulting in severe injury, loss of mobility, and decreased independence [1]. Effective fall detection systems are essential for timely intervention, yet current models face several limitations in complex indoor environments. Traditional approaches struggle with challenges such as occlusion, cluttered scenes, and variable lighting, leading to high rates of false detections or missed events. These limitations underscore the need for a more robust and adaptable system for real-world deployment.

This paper introduces a customized YOLOv5 model, enhanced with Dynamic Atrous Convolution (DAC) and Self-Calibrated Convolution (SCConv). The DAC component dynamically adjusts the receptive field, enabling the model to detect people at multiple scales, while SCConv provides self-calibration to improve robustness against occlusion and lighting variations. Together, DAC and SCConv help the model capture diverse spatial features in cluttered indoor environments.

In this system, YOLOv5 is used for person detection, and the detected regions are processed by MediaPipe, which performs pose estimation and extracts keypoints to analyze relative positions of the head, torso, and hips. This integrated approach aids in identifying potential falls accurately. The proposed model is deployed using a Kinect V2 camera, allowing for real-time, deep vision deployment in various indoor settings.

The primary contribution of this paper is the enhancement of the YOLOv5 model through the integration of Dual Attention Convolution (DAC) and Self-Calibrated Convolution (SCConv) modules. This upgraded YOLOv5 model is further combined with MediaPipe's keypoint detection to accurately identify true fall events by analyzing descent velocity and bounding box aspect ratio, triggering an alarm upon confirmed detection. The result is a robust, multi-modal fall detection system with real-world applicability, designed to reliably detect falls even in challenging conditions.

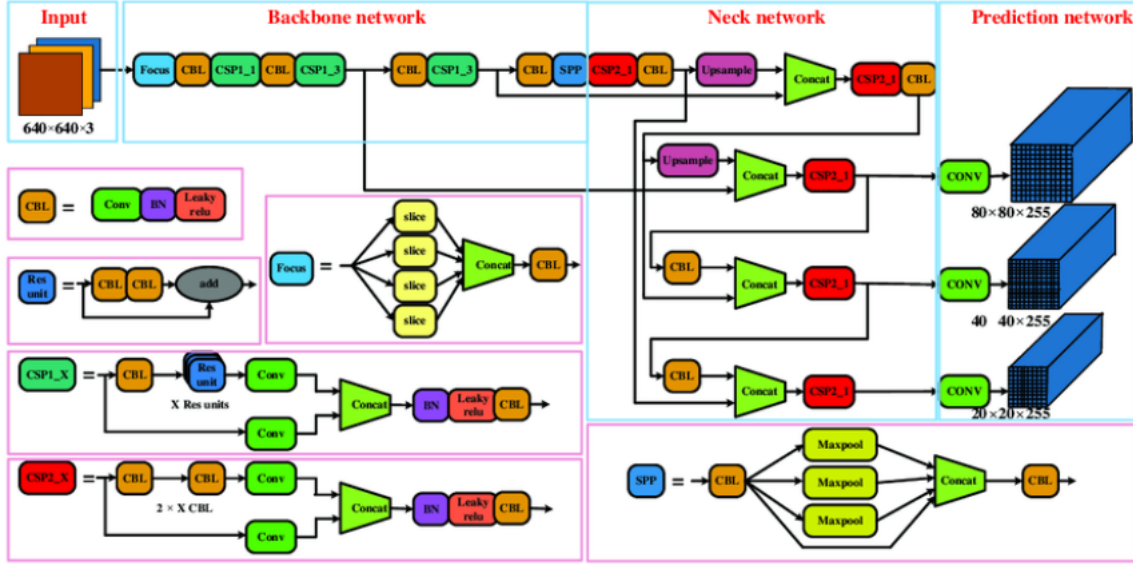


Figure 1: Yolo v5 component structure consisting of the backbone with Focus layer responsible for splitting the image into multiple slices, Cross Stage Partial (CSP) and Convolutional-BatchNorm-Leaky (CBL) block modules to enhance feature extraction, neck and prediction network for object detection [5].

2 RELATED THEORIES

2.1 Introduction to YOLO v5

YOLOv5s is a preferred choice for real-time applications due to its efficiency, offering a balance between detection accuracy and computational efficiency [2]. The YOLOv5 architecture consists of three main components: the Backbone, the Neck, and the Prediction network as shown in Figure 1.

Each part plays a crucial role in achieving accurate and fast object detection [3], [4]. The Backbone network extracts features from the input image, gradually building up a high-dimensional representation that captures essential details like textures, edges, and object parts [5]. YOLOv5 uses a CSPDarknet53-based backbone that includes several Convolutional-BatchNorm-Leaky (CBL) blocks, Cross Stage Partial (CSP) networks, and Spatial Pyramid Pooling (SPP) modules. Key components include the Focus layer, which splits the input image into multiple slices to reduce computation, effectively downsampling it to focus on essential information while preserving spatial details. CSP modules, such as CSP1_3 and CSP2_3, enhance feature extraction by creating residual paths and dividing the input across multiple layers, enabling better gradient flow and reducing computation by allowing feature reuse, which enhances performance without significantly increasing model complexity.

The CBL block (Convolutional-BatchNorm-Leaky) which combines a convolutional layer, batch normalization, and a leaky ReLU activation function. It plays a key role in feature extraction and representation. Additionally, the SPP (Spatial Pyramid Pooling) module increases the receptive field by applying multiple pooling layers with different kernel sizes, helping the model capture more contextual information, which is particularly useful for detecting

objects at varying scales. However, the Backbone network has limitations in handling occlusions and complex scenes with overlapping objects. Its fixed receptive field size can make it difficult for YOLOv5 to capture objects at different depths or accurately detect them in cluttered environments.

The Neck network helps aggregate features from different stages of the Backbone and passes them to the Prediction network. YOLOv5's Neck is designed with feature pyramids and upsampling layers to strengthen multiscale feature extraction. Important components include upsampling layers that increase the spatial resolution of the feature maps, allowing the model to retrieve detailed information about smaller objects. Skip connections and concatenation layers combine feature maps from different depths, ensuring that information from both shallow (fine details) and deep layers (semantic context) is available for final prediction.

The Prediction network is responsible for the final object detection and classification. It consists of three detection layers, each focused on different feature map scales: large-scale detection for small objects, medium-scale detection for medium-sized objects, and small-scale detection for large objects. Each detection layer produces bounding boxes along with class scores for detected objects at different levels of detail.

2.2 Mediapipe for pose estimation

MediaPipe is a versatile framework developed by Google, designed for real-time machine learning applications in areas like hand tracking, pose estimation, and face detection. Its architecture consists of a modular pipeline that processes input data through a series of interconnected nodes, each dedicated to a specific function (e.g., image preprocessing, landmark detection). The framework uses lightweight neural networks optimized for efficient computation,

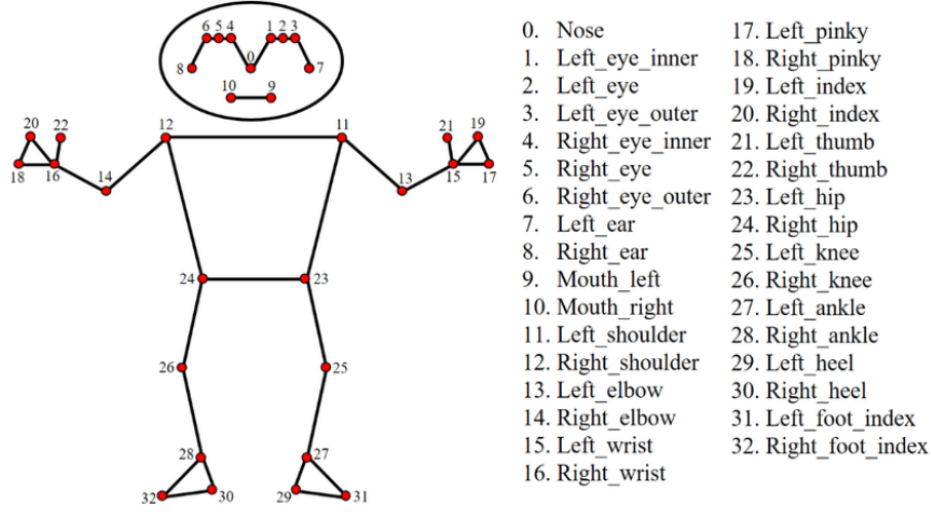


Figure 2: Mediapipe 33 keypoints that represent different parts of the human body [7].

allowing it to run smoothly on various devices, including mobile and edge devices [6].

In the context of pose estimation, MediaPipe provides 33 keypoints [7] representing different parts of the human body, as shown in Figure 2.

These keypoints include locations on the head, shoulders, elbows, wrists, hips, knees, ankles, and specific parts of the face, hands, and feet. Examples include the nose (0), shoulders (11, 12), elbows (13, 14), wrists (15, 16), hips (23, 24), knees (25, 26), and ankles (27, 28), among others. The comprehensive set of keypoints allows for precise tracking of body movements and detailed understanding of posture. By analyzing the relative positions of these keypoints (e.g., comparing the head and hip heights), we can determine if an individual has fallen. Specifically, we can detect a fall if the head and torso keypoints suddenly approach ground level or display unusual angles, which is critical for distinguishing a fall from other non-harmful postures (e.g., sitting or lying down). This integration of MediaPipe’s keypoints ensures that our system accurately verifies a fall, reducing false positives and increasing the reliability of the detection model.

2.3 Limitations of Current Approaches

Tingting Chen et al. [8] addressed the challenges of elderly fall detection by incorporating asymmetric convolution blocks (ACB) and a spatial attention mechanism into YOLOv5s. The ACB modules were designed to improve feature extraction by combining horizontal, vertical, position, and rotation features, while the spatial attention mechanism emphasized relevant spatial areas in the network, enhancing the detection of features crucial for fall detection.

However, the reliance on fixed-scale features and limited adaptability in occluded or cluttered scenes limits the robustness of this approach. Specifically, the fixed convolution structures in ACB may not dynamically adjust to varying object sizes or scales in real-world scenarios, potentially leading to inaccuracies when objects

are partially occluded or small in size. Similarly, while the spatial attention mechanism enhances feature focus, it lacks dynamic adjustments for multi-scale contextual awareness.

While MediaPipe provides an efficient and lightweight framework for real-time keypoint detection, its performance is constrained by several limitations. As noted by Docekal et al. [9], MediaPipe’s dependency on the visibility of body parts leads to degraded accuracy when keypoints are occluded or outside the frame. Furthermore, the lack of depth context or 3D spatial understanding can result in incorrect posture estimations in complex scenarios, particularly in applications requiring precise spatial interpretation. Additionally, the pretrained models in MediaPipe may struggle to adapt to diverse datasets or accommodate non-standard body proportions, limiting its applicability in heterogeneous real-world settings.

To address these challenges, our methodology employs a confidence-based fusion of YOLOv5’s bounding box detections with MediaPipe’s keypoint estimations. By leveraging the complementary strengths of both frameworks, this approach mitigates the limitations of MediaPipe. For instance, YOLOv5’s bounding boxes provide robust spatial localization, guiding MediaPipe’s pose estimation to reduce errors caused by occlusions or missing keypoints. Weighted Keypoint Adjustment ensures that bounding boxes with high confidence are prioritized, filtering out inaccurately detected keypoints. Furthermore, IoU-Enhanced Keypoint Confidence refines pose estimations by retaining only those keypoints that align closely with YOLOv5’s bounding boxes, effectively compensating for MediaPipe’s lack of depth awareness and improving reliability in complex scenarios. This integration enhances the robustness of our fall detection system, overcoming the constraints inherent in using MediaPipe or YOLOv5 independently.

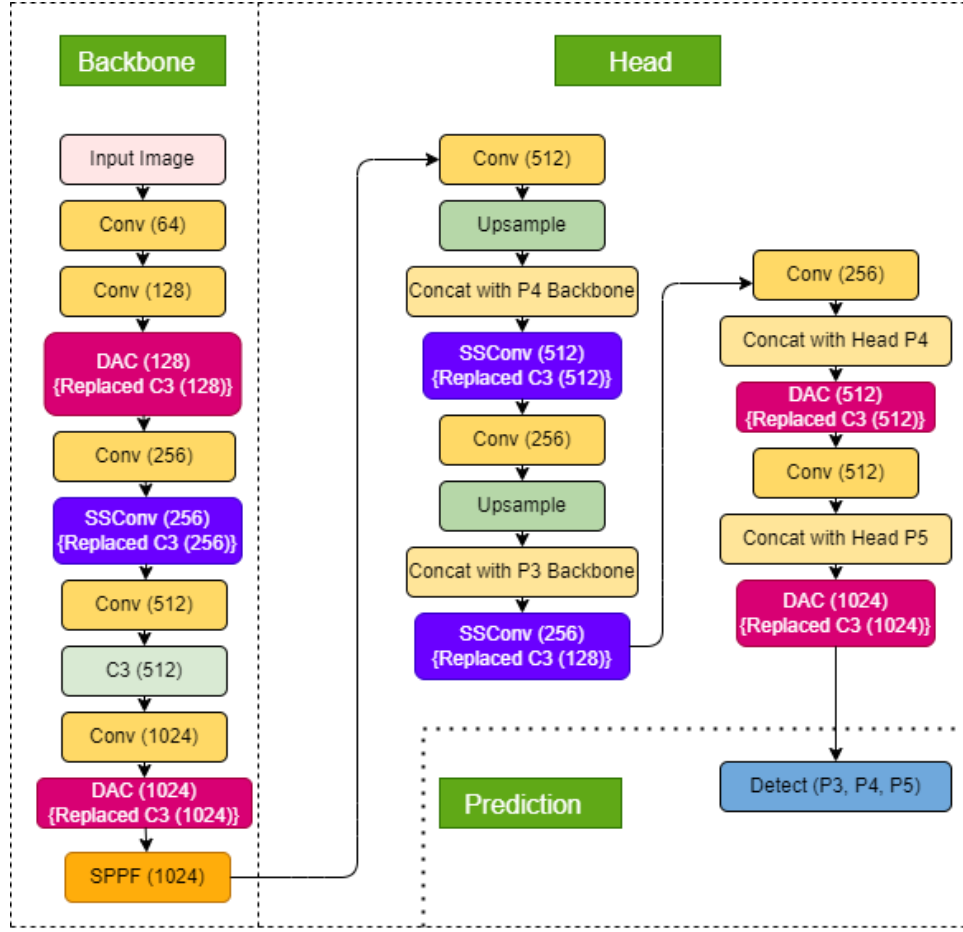


Figure 3: Improved YOLOv5 which replaces several C3 modules (representing the Cross Stage Partial (CSP) structure) located in the head and backbone networks.

3 THE PROPOSED METHOD

The proposed architecture modifies the YOLOv5 model as seen in Figure 3 by integrating Dynamic Atrous Convolution (DAC) and Self-Calibrated Convolution (SSConv) blocks into both the backbone and head, enhancing feature extraction and adaptability. Retaining YOLOv5’s backbone-neck-head structure, it strategically replaces certain C3 layers with DAC and SSConv to introduce multi-scale contextual awareness and spatial refinement. DAC modules dynamically adjust the receptive field to suit various object scales, while SSConv enhances spatial awareness by refining focus on critical regions. The backbone concludes with an SPPF module to fuse multi-scale features. In the head, upsampling and concatenation operations combine features from different scales, further refined by DAC and SSConv. This design maintains YOLOv5’s efficiency while improving its robustness.

3.1 Dynamic Atrous Convolution (DAC)

Dynamic Atrous Convolution (DAC) enhances YOLOv5’s adaptability by allowing the receptive field to adjust dynamically to different object scales, which is crucial for detecting objects in complex

scenes. DAC builds on standard atrous (dilated) convolution [10], where the dilation rate r determines the spacing of kernel elements, effectively expanding the receptive field without increasing the number of parameters. The concept of atrous convolution was first introduced in DeepLab [11] and is sometimes referred to as the atrous convolution layer. Its purpose is to generate high-resolution feature maps by capturing a larger contextual area while keeping the parameter count consistent. This approach, shown to be effective for semantic segmentation, has also been incorporated into prominent object detectors like SSD and R-FCN to improve both speed and accuracy.

The output of a DAC operation for an input feature map x and kernel w is given by:

$$y_i = \sum_{k=0}^n x[i + r \cdot k] \cdot w[k], \quad (1)$$

where i indexes spatial locations in the feature map, k indexes the kernel elements, and r is adjusted based on the scale of the object being detected. DAC dynamically changes r according to spatial context as shown in Figure 4, enabling the model to capture fine details and broader contextual information as needed. This

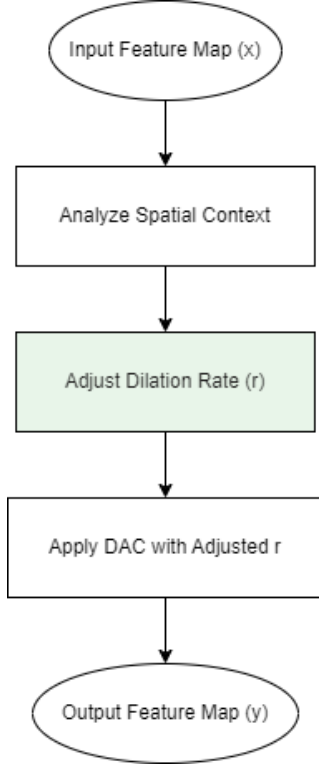


Figure 4: Output of a DAC operation for an input feature map x with the Dilation Rate dynamically changing according to the spatial context of the Input.

adaptability is particularly valuable in scenes with objects at varying depths and sizes, helping the model effectively handle cluttered or occluded environments.

In practice, DAC stacks multiple atrous convolution layers with different dilation rates, allowing the network to process multi-scale features in a single layer. This multi-scale processing improves the model’s sensitivity to objects of various sizes without requiring separate layers for each scale.

3.2 Self-Calibrated Convolution (SCConv)

Self-Calibrated Convolution (SCConv) is a technique designed to improve spatial feature refinement by enhancing the model’s ability to focus on relevant regions of an input [10]. Unlike standard convolution, which applies a fixed kernel across the input, SCConv introduces a self-calibration process that adjusts the feature map’s values based on the contextual information of neighboring elements, allowing for adaptive feature extraction. This dynamic adjustment enhances the model’s response to significant spatial patterns while suppressing less informative areas, making SCConv particularly useful in complex scenes where object boundaries may be unclear or partially occluded.

SCConv is composed of two main branches: the context branch and the information branch [12]. For an input feature map x of spatial size $H \times W$, SCConv first splits the input into two separate branches as shown in Figure 5.

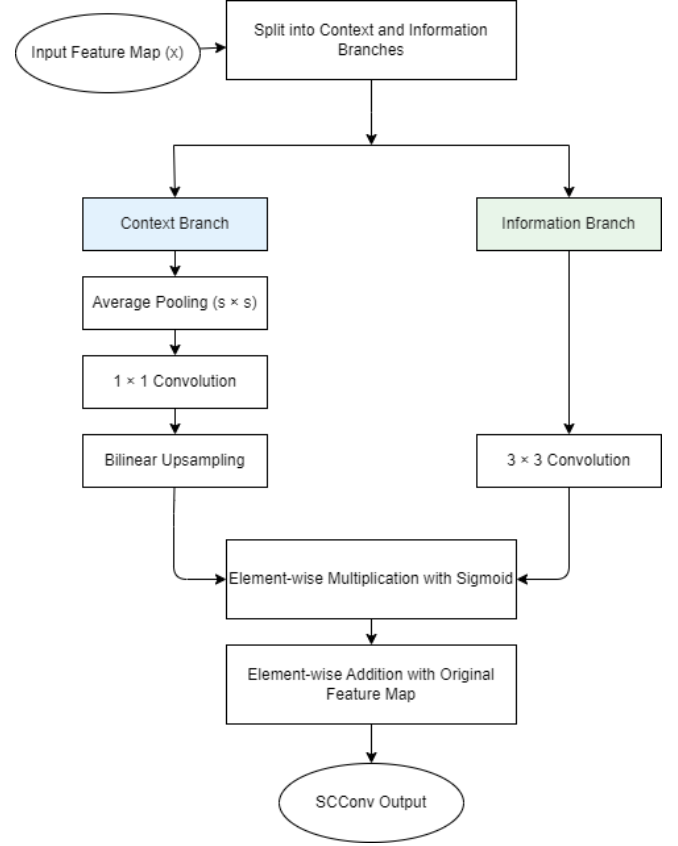


Figure 5: SCConv is composed of context and information branches after Input map feature split. The context branch does an average pooling operation for capturing coarse-grained contextual information and the information branch which directly processes input to preserve local details and enhance fine-grained features.

In the context branch, an average pooling operation with a pre-defined window size $s \times s$ is applied to the feature map to capture coarse-grained contextual information:

$$x_{context} = AvgPool(x, s) \quad (2)$$

where $x_{context}$ is the downsampled feature map that retains essential spatial information over a larger receptive field. After pooling, a 1×1 convolution is applied to the pooled output to reduce dimensionality and generate a compact representation of the context. This context representation is then upsampled back to the original resolution using bilinear interpolation to match the feature map size.

In parallel, the information branch directly processes the input feature map through a 3×3 convolution. This operation preserves local details and enhances fine-grained features by generating a response map that focuses on spatially dense information:

$$x_{info} = Conv_{3 \times 3}(x_{info} = Conv_{3 \times 3}(\quad (3)$$

Next, the output of the information branch, x_{info} , is multiplied element-wise with the upsampled context feature map $x_{context}$ to

form a self-calibrated response map:

$$y = x_{info} \times \partial(x_{context}) \quad (4)$$

where ∂ denotes the sigmoid activation function, which normalizes the context features to a range between 0 and 1. This normalization helps highlight the important spatial regions by weighing the local information in x_{info} based on the broader context from $x_{context}$. The self-calibrated response y is then combined with the original input feature map via an element-wise addition to produce the final SCCConv output, which can be written as:

$$y_{SCCConv} = x + y \quad (5)$$

This addition step allows the network to retain the original feature representation while integrating the calibrated features.

3.3 Confidence-Based Fusion of YOLOv5 and MediaPipe Keypoints for Fall Detection

To improve the accuracy of fall detection in complex environments, we adopt a confidence-based fusion method that integrates YOLOv5's bounding box detections with MediaPipe's keypoint estimations. This approach leverages YOLOv5's robust object detection capabilities alongside MediaPipe's precise pose estimation to refine the decision-making process. By combining these complementary strengths, the system ensures more reliable fall detection through weighted adjustments and IoU-based validation.

The process begins with analyzing YOLOv5's confidence scores for each detected bounding box to establish the reliability of detections. For each keypoint identified by MediaPipe, an adjusted confidence score is computed by combining YOLOv5's detection confidence with MediaPipe's keypoint confidence. The adjusted confidence c_i for a keypoint is given by the equation YOLO:

$$c_i = w \times c_{YOLO} + (1 - w) \times c_{MediaPipe} \quad (6)$$

where c_{YOLO} represents YOLOv5's confidence score for the corresponding bounding box, $c_{MediaPipe}$ represents MediaPipe's confidence for the keypoint, and w is a weighting factor that prioritizes YOLOv5's detection when its confidence is high. This weighted adjustment helps filter out unreliable keypoints and ensures that detections with stronger evidence are given precedence.

$$IoU = \frac{B_{YOLO} \cap B_{MP}}{B_{YOLO} \cup B_{MP}} \quad (7)$$

where B_{YOLO} is the bounding box detected by YOLOv5 and B_{MP} is the bounding box encompassing the keypoints detected by MediaPipe. This calculation evaluates the overlap between the two bounding boxes. If the IoU score exceeds a predefined threshold, the corresponding keypoints are deemed reliable and retained for further analysis. Keypoints with an IoU below the threshold are down-weighted to reduce the likelihood of false positives. By integrating these methods, the system leverages the strengths of both YOLOv5 and MediaPipe to refine keypoint detections and improve the robustness of the overall fall detection process.

Fall detection is evaluated using two key metrics. First, the vertical velocity of the chest and hip keypoint is monitored by analyzing its positional change across consecutive frames. If the velocity of decent exceeds a predefined threshold while the individual's height is decreasing, it indicates a rapid downward movement suggesting a potential fall. This step ensures that only significant downward

movement is considered, filtering out minor changes in posture or small motions that do not represent falls. Second, the aspect ratio of the bounding box, defined as its width divided by its height, is tracked. A significant shift from a tall, narrow shape (indicating standing) to a wide, flat shape (indicating lying down) whose ratio will be greater than 1, provides another critical indicator of a fall. Together, these metrics enable robust and accurate fall detection even in challenging scenarios as shown in Figure 6.

4 EXPERIMENT RESULTS AND ANALYSIS

4.1 Dataset and Image Processing

In our experimental setup, we used a dataset consisting of a total of 1,581 images sourced from [13], [14], [15]. These images were preprocessed and augmented using Roboflow [16] to enhance the robustness of the model and to introduce variability, ensuring better generalization.

Preprocessing involved applying auto-orientation to align all images consistently. Each image was then resized by stretching it to a uniform resolution of 640x640 pixels, ensuring compatibility with the model input requirements.

Data augmentation was applied to create multiple variations of each image and simulate different conditions. For each training example, three output variations were generated. The augmentation techniques included horizontal flipping, rotation, grayscale transformation, exposure adjustment, and noise addition. Horizontal flipping was used to improve model invariance to orientation, while rotation within a range of -15° to $+15^\circ$ introduced variations in image perspective. Grayscale conversion was applied to 15% of the images to make the model more adaptable to varying color intensities. Exposure adjustments ranging between -10% and $+10\%$ were applied to simulate different lighting conditions, while random noise was added to up to 0.1% of the pixels, which helps the model handle noisy or imperfect input data more effectively.

The dataset was split into training, validation, and testing sets to support training and performance evaluation in the ratio of 8:1:1. This setup aimed to maximize the utility of the dataset by providing balanced and diverse samples across training, validation, and testing stages.

4.2 Evaluation Metrics

To evaluate the performance of our fall detection model, we use three primary indicators: precision, recall, and mean average precision (mAP). Each of these metrics provides insights into the model's ability to detect falls accurately and minimize false detections.

Precision P is a measure of the model's accuracy in identifying true falls among all detections made. Precision calculates the ratio of true positives (correctly detected falls) to the sum of true positives and false positives (incorrect fall detections). A high precision value indicates that the model makes few false detections, meaning it rarely identifies a non-fall event as a fall. However, precision alone does not consider missed fall detections. Precision is calculated as follows:

$$P = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

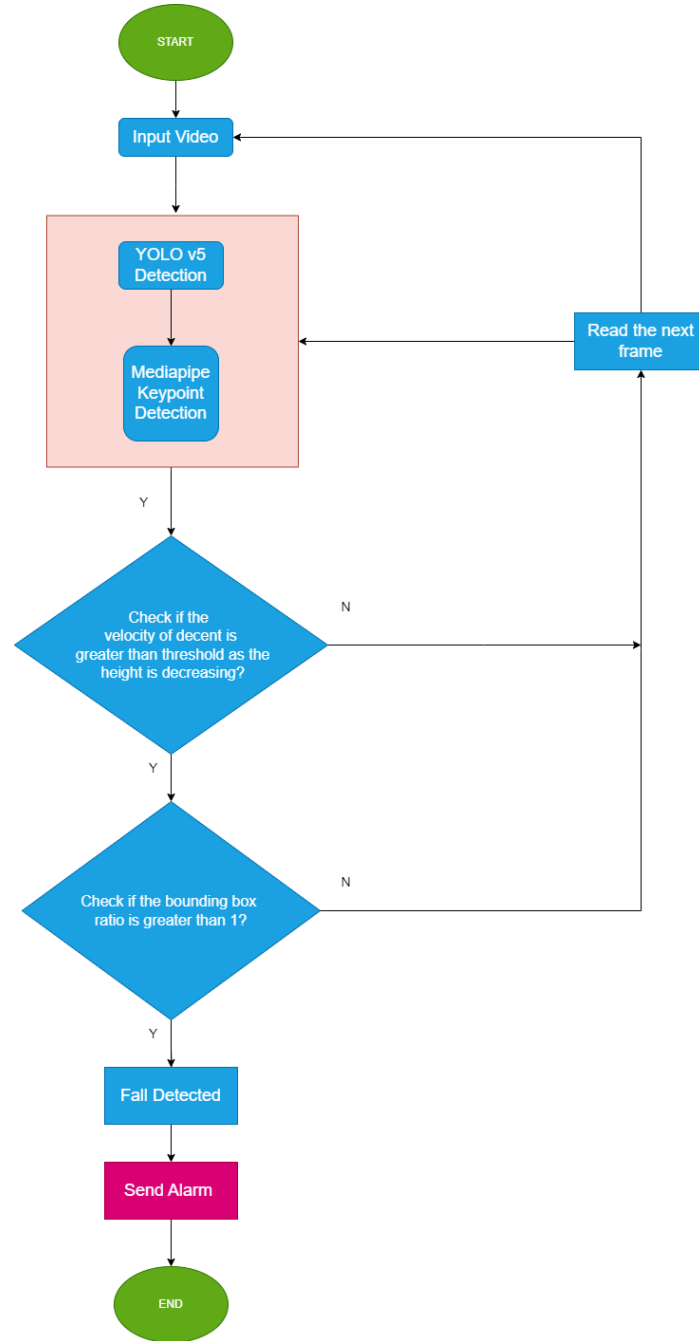


Figure 6: Fall detection flow chart

Recall R reflects the model's sensitivity to actual fall events. It measures the proportion of true fall detections among all actual falls present in the data. A high recall means that the model is effectively capturing most actual falls, reducing the risk of missed events. However, high recall might come at the expense of precision, as the model might detect some non-falls as falls to ensure it doesn't

miss any. Recall is calculated as follows:

$$R = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

Mean Average Precision (mAP) is an aggregated measure of the model's precision across varying recall levels. It captures both precision and recall by averaging the precision scores at multiple points along the recall curve, resulting in a single performance

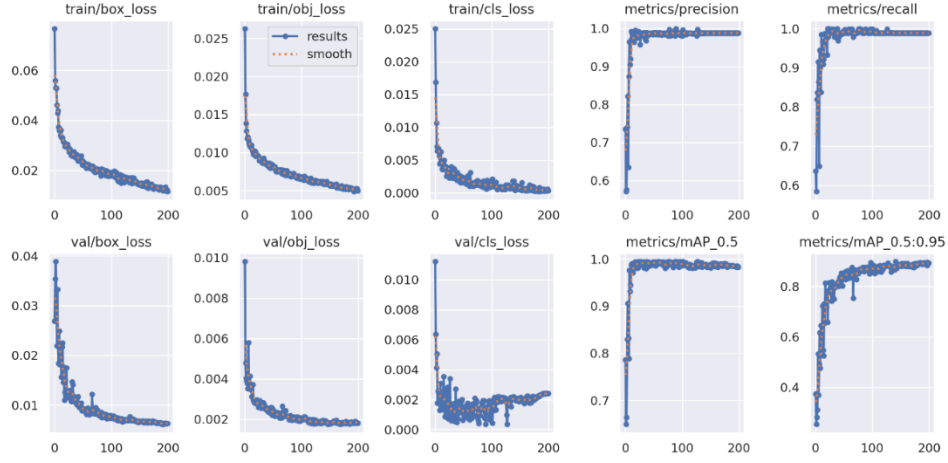


Figure 7: Model performance after training with loss curves, precision, recall and mAP metrics.

Table 1: Environment Settings for the Trained Model

Specification	Details
Operating System	Ubuntu 18.04
CPU	3.2 GHz AMD Ryzen 7
GPU	NVIDIA GeForce RTX 3070
RAM	16 GB DDR4
Language	English
DL Framework	PyTorch 1.13.1

metric. mAP is especially valuable in object detection tasks because it balances the trade-off between detecting all instances of the target class (high recall) and minimizing incorrect detections (high precision). A higher mAP score indicates a model that performs well in identifying true falls with minimal false alarms across different thresholds.

4.3 Result Analysis

The results of our fall detection model training are illustrated in Figure 7, which shows various metrics and loss curves over 200 training epochs.

These metrics include box loss, object loss, classification loss, precision, recall, and mean average precision (mAP) at IoU thresholds of 0.5 and 0.5:0.95. The Improved YOLOv5 model demonstrated achievements in terms of mAP, showcasing the effectiveness of the training process. The model achieved a mAP@0.5 of 98.6% and a mAP@0.5:0.95 of 89.6% on the validation dataset. Information about the experimental environment is provided in Tables 1 and 2.

The training and validation box loss, object loss, and classification loss curves show a consistent decrease as training progresses. Box loss, which measures the error in the localization of bounding boxes, starts high and gradually decreases to below 0.02, indicating that the model is learning to accurately localize detected objects. Object loss, reflecting the model’s confidence in object presence, also decreases steadily, converging to a minimal value below 0.01 by

Table 2: Hyperparameter Options and Values

Hyperparameter Option	Value
Image size	640 × 640
Initial learning rate	0.01
Learning rate float	0.01
Optimizer	SGD
Momentum	0.937
Weight decay	0.0005
Batch size	16
Epochs	200

the end of training. Classification loss, which captures the accuracy of class predictions, reaches a low value as well, suggesting the model is accurately identifying falls. Figure 8 shows validation results on the dataset, with potential fall and non-fall scenarios detected accurately.

The precision and recall metrics exhibit high values, both converging to around 1.0 by the end of training. This indicates that the model is effectively identifying true fall instances with minimal false positives (high precision) and successfully capturing most of the actual falls (high recall). High precision and recall together indicate that the model performs well in both correctly identifying falls and minimizing missed detections.

The mAP values, which provide a more comprehensive measure of detection performance across multiple IoU thresholds, show stable high values as training progresses. mAP at 0.5 reaches close to 1.0, suggesting that the model performs accurately when considering only moderate overlaps between predicted and ground truth bounding boxes. The mAP at 0.5:0.95, which averages over stricter IoU thresholds, converges around 0.8 to 0.9, indicating that the model maintains good detection performance even under more challenging evaluation conditions. This high mAP value demonstrates the model’s robustness and ability to maintain reliable fall detection across varying degrees of overlap requirements.



Figure 8: Validation results on the dataset, with potential fall and non-fall scenarios detected accurately

4.4 Ablation Study

In this section, we evaluate the performance impact of various modifications introduced to the YOLOv5s architecture. The baseline model, labeled as "YOLOv5," serves as a comparison point for the additions we made as shown in Figure 9.

Table 3: Performance Comparison of modules

Model	mAP score (%)
Yolov5	93.8
DAC	94.7
SCConv	96.2
DAC+SCConv	98.6

We progressively added the Dynamic Atrous Convolution (DAC) module, Self-Calibrated Convolution (SCConv) module, and a combination of both DAC+SCConv to assess their effects on mean Average Precision (mAP) further tabulated in Table 3.

Baseline YOLOv5: The original YOLOv5 model achieved an mAP score of 0.938. This represents the starting point before any architectural modifications were introduced.

YOLOv5 + DAC: Incorporating the DAC module, which enhances multi-scale feature extraction, increased the mAP score to 0.947. This improvement indicates that the DAC module helps the model capture features at different scales, improving object detection accuracy.

YOLOv5 + SCConv: Replacing standard convolutions with the SCConv module, which refines spatial information in the feature maps, further boosted the mAP to 0.962. The SCConv module's spatial refinement proved beneficial, allowing the model to distinguish object boundaries and spatial details more effectively.

YOLOv5 + DAC + SCConv: Finally, combining both DAC and SCConv modules led to a significant improvement, resulting in an mAP score of 0.986. This combination allows the model to leverage both multi-scale feature extraction and enhanced spatial refinement, achieving the best performance among all configurations.

These results highlight the effectiveness of each module independently, as well as the synergy achieved by combining DAC and SCConv. The addition of DAC and SCConv modules enables the model to capture a richer set of features, leading to a considerable performance boost in object detection tasks.

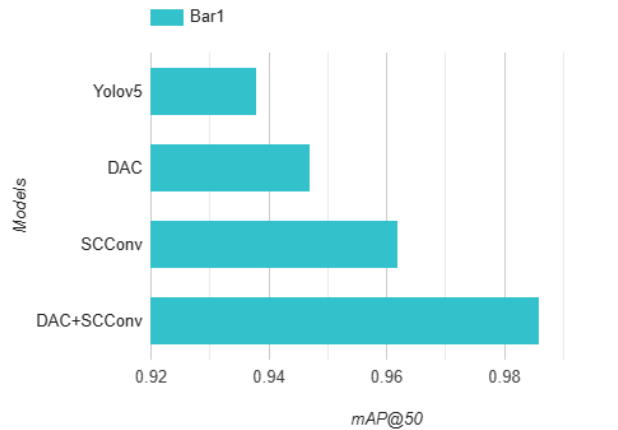


Figure 9: Performance comparison of the SCConv and DAC modifications to the baseline YOLOv5 model



Figure 10: Fall scenario captured after accurately detecting the person's body orientation and keypoints.

Table 4: Comparison of different object detection frameworks

Model	mAP score (%)
YOLOv5	93.8
YOLOv7	91.5
YOLOv8	92.8
OURS	98.6

4.5 Real world scenario experiment and comparison

In the final part of our evaluation, we demonstrate the fall detection system in action using the Kinect v2 camera. As shown in Figure 10, a fall scenario is captured where the system accurately detects the person's body orientation and keypoints using MediaPipe. Once the system identified a fall based on the descent velocity of the chest and the bounding box ratio, an alarm was automatically triggered to alert caregivers or monitoring personnel.

A comparative evaluation as shown in table 4 was conducted between our enhanced YOLOv5 model and recent object detection frameworks, including YOLOv8 (92.8% mAP), YOLOv7 (91.5% mAP), and the original YOLOv5 (93.8% mAP). Our modified YOLOv5, incorporating DAC and SCConv modules, achieved a significant improvement with a mean Average Precision (mAP) of 98.6%, outperforming the other models and demonstrating the effectiveness of our enhancements for reliable fall detection.

Table 5 presents a comparative evaluation of recent approaches for fall detection on the UR-FALL dataset, showcasing their respective methodologies and achieved mAP scores. The table highlights the performance of various enhancements to the YOLO architecture. For instance, Kan et al. utilized the CGNSYOLO method, achieving an mAP of 91.3%, while Luo et al. integrated GCC into YOLOv5s,

attaining 91.9%. Chen et al. further improved performance by incorporating ACB into YOLOv5, resulting in an mAP of 97.2%. In comparison, making use of the UR-FALL dataset as used in the compared referenced papers, our approach, which integrates Dynamic Atrous Convolution (DAC) and Self-Calibrated Convolution (SCConv) into YOLOv5, outperformed these methods with an mAP score of 98.6%.

5 CONCLUSION

In this paper, we presented a robust and innovative fall detection system tailored for elderly care, leveraging a customized YOLOv5 model enhanced with Dynamic Atrous Convolution (DAC) and Self-Calibrated Convolution (SCConv). Our approach effectively addresses the challenges posed by complex indoor environments, such as occlusion, clutter, and variable lighting, resulting in a significant reduction in false detection rates. The integration of YOLOv5 for person detection and MediaPipe for pose estimation allows for accurate monitoring of key body positions, culminating in a mean Average Precision (mAP) of 98.6%.

The key contributions of this work include the integration of DAC and SCConv modules into YOLOv5's backbone and head, which enhanced the model's ability to perform multi-scale feature extraction and spatial refinement, effectively improving detection accuracy. Additionally, the confidence-based fusion of YOLOv5's bounding box detections and MediaPipe's keypoint estimations introduced a robust mechanism for fall detection by leveraging metrics like descent velocity and bounding box aspect ratio.

The findings of this study not only advance the state of fall detection technology but also provide a practical framework for real-world applications, enhancing the safety and responsiveness of elderly care systems.

For future work, we aim to further optimize the model by reducing the number of parameters to ensure faster responsiveness without compromising detection accuracy.

6 AUTHORS AND AFFILIATIONS

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Table 5: Comparative evaluation recent related approaches

Reference	Methodology	mAP score (%)
Kan et al. [18]	Designed CGNSYOLO	91.3
Luo et al [19]	YOLOv5s+GCC	91.9
Chen et al	YOLOV5+ACB	97.2
OURS	YOLOv5+DAC+SCConv	98.6

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